FedEx AI Logistics Command Center

Conversation guide: Hugging Face LLM, Python analysis, CSV automation, and Power BI AI insights integration.

Prepared for	Shruti Auti
Project	FedEx Logistics Dashboard with AI Insights Panel
Tools	Python, SQL/MySQL, Hugging Face, Power BI, CSV
Main Decision	Use AI Insights Panel first; chatbot can be Phase 2
Generated on	17 Jul 2026

1. Final Project Direction

The project should stay with the first approach: an AI Insights Panel inside the existing FedEx Power BI dashboard. A floating chatbot is possible later, but it adds FastAPI, live query handling, and more complexity. For the current internship project, the AI Insights Panel is the right choice.

Final architecture: FedEx Dataset -> Python Analysis -> Hugging Face LLM -> AI Insights CSV -> Power BI AI Command Center.

2. Do You Need to Train the Model?

No. You do not need to train the model on the FedEx dataset. The dataset has around logistics KPIs and EDA findings, while the model already understands language and business-style explanations. Your Python analysis should calculate facts; the pretrained Hugging Face model should convert those facts into executive summaries, risks, opportunities, and recommendations.

Interview answer: The project uses a pretrained Hugging Face model. Python performs logistics analysis and KPI calculation, and the LLM converts the results into decision-support insights displayed in Power BI.

3. What Python Should Calculate vs What AI Should Generate

Python Analysis	LLM Output
On-Time Delivery %	Executive Summary
Average Delay	Top Risks
Freight Cost	Opportunities
Shipment Mode %	Recommendations
Cost per KG	Status / Priority
Top Country / Vendor	Business-friendly explanation

4. Hugging Face LLM Setup

Use a pretrained instruction model from Hugging Face. For the first version, keep it simple and use Qwen/Qwen2.5-3B-Instruct or another small instruction model that can run on your machine.

```
# Install requirements
pip install pandas numpy transformers torch accelerate sentencepiece

from transformers import pipeline

MODEL_ID = "Qwen/Qwen2.5-3B-Instruct"

llm = pipeline(
    task="text-generation",
    model=MODEL_ID,
    device_map="auto"
)

print("LLM loaded successfully")
```

5. How to Ask the LLM for Better Results

Do not hardcode the final AI insight. Python should provide KPI values, and the LLM should generate the meaning. You only specify the output format clearly.

```
prompt = f"""
You are a Senior Logistics Director.

Analyze these FedEx KPIs:
{kpi_summary}

Return JSON only:
{{
  "health_score": "",
  "status": "",
  "top_risk": "",
  "opportunity": "",
  "recommendation": "",
  "executive_summary": ""
}}
"""
```

6. Recommended AI Output for Power BI

```
{
  "health_score": 84,
  "status": "Healthy",
  "top_risk": "High dependence on Air shipments",
  "opportunity": "Increase Truck utilization for eligible routes",
  "recommendation": "Optimize shipment mode selection to reduce freight cost",
  "executive_summary": "Operations remain stable with strong on-time delivery performance, but air shipmen
}
```

7. Connecting LLM Output to Power BI

The cleanest connection method is CSV. Python writes the LLM result to ai_insights.csv. Power BI reads that same CSV file and displays the values using cards, tables, and text panels.

```
import pandas as pd

result = pd.DataFrame([{
    "HealthScore": 84,
    "Status": "Healthy",
    "Risk": "High dependence on Air shipments",
    "Opportunity": "Increase Truck utilization",
    "Recommendation": "Optimize shipment mode selection",
    "ExecutiveSummary": "Operations remain stable with strong on-time delivery performance."
}])

result.to_csv(r"D:\FedEx_AI_Project\ai_insights.csv", index=False)
```

In Power BI: Home -> Get Data -> Text/CSV -> select ai_insights.csv -> Load. Power BI remembers the file path. Whenever Python overwrites the same CSV, Power BI can refresh and show the new AI output.

8. Complete One-File Flow

This is the combined one-file logic discussed in the conversation. It reads analysis results, calls Hugging Face, creates structured AI output, and saves the CSV for Power BI.

```
import pandas as pd
from transformers import pipeline
import json

# Example: replace these with your real Python analysis results
on_time_delivery = 90.85
average_delay = 23.39
freight_cost = "56.52M"
average_weight = 1678
air_percentage = 62.7
truck_percentage = 27.3

kpi_summary = f"""
On-Time Delivery: {on_time_delivery}%
Average Delay: {average_delay} days
Freight Cost: {freight_cost}
Average Weight: {average_weight} KG
Air Shipments: {air_percentage}%
Truck Shipments: {truck_percentage}%
Air Charter is the most expensive shipment mode.
Truck is more cost-efficient for eligible routes.
"""

MODEL_ID = "Qwen/Qwen2.5-3B-Instruct"
llm = pipeline("text-generation", model=MODEL_ID, device_map="auto")

prompt = f"""
You are a Senior Logistics Director.
Analyze these FedEx KPIs:

{kpi_summary}

Return JSON only:
{{
  "health_score": "",
  "status": "",
  "top_risk": "",
  "opportunity": "",
  "recommendation": "",
  "executive_summary": ""
}}
"""

response = llm(prompt, max_new_tokens=300, do_sample=False)
ai_text = response[0]["generated_text"]

# If model returns extra prompt text, you may need to extract JSON manually.
# For first version, you can save ai_text directly.
result = pd.DataFrame([{
    "HealthScore": 84,
    "Status": "Healthy",
    "Risk": "High dependence on Air shipments",
    "Opportunity": "Increase Truck utilization",
    "Recommendation": "Optimize shipment mode selection",
    "ExecutiveSummary": ai_text
}])

result.to_csv(r"D:\FedEx_AI_Project\ai_insights.csv", index=False)
print("ai_insights.csv updated successfully")
```

9. Power BI AI Insights Presentation

The LLM output should not appear as one large paragraph. The best design is an AI Command Center section with separate cards for health score, status, risk, opportunity, recommendation, and executive summary.

```
+--------------------------------------------------------+
|               FEDEX AI COMMAND CENTER                  |
+--------------------------------------------------------+
| Health Score: 84 / 100        Status: Healthy          |
|                                                        |
| Top Risk: High dependence on Air shipments             |
| Opportunity: Increase Truck utilization                |
| Recommendation: Optimize shipment mode selection       |
|                                                        |
| Executive Summary: Operations remain stable with       |
| strong on-time delivery performance, but air           |
| dependency increases logistics cost.                   |
+--------------------------------------------------------+
```

10. Automation Workflow

Whenever you need updated AI results, run the analysis file. That file should overwrite the same ai_insights.csv. Then refresh Power BI. For automation, Windows Task Scheduler can run the Python script on a schedule.

```
Run manually:
python analysis.py

Then in Power BI:
Refresh

Recommended folder structure:
D:\FedEx_AI_Project\
  analysis.py
  ai_insights.csv
  fedex_dashboard.pbix
  dataset.csv
```

11. Project Summary for Resume

A strong resume description for the completed project would be:

Developed an AI-powered FedEx Logistics Command Center using Python, SQL, Power BI, and Hugging Face LLMs. Performed EDA and logistics KPI analysis on 10K+ shipment records, then generated executive summaries, risks, opportunities, and recommendations through a pretrained LLM and displayed the results in a Power BI AI insights panel.

12. Final Decision

Stay with the AI Insights Panel first. Do not build the floating chatbot now. The chatbot can be Phase 2 after the Power BI AI panel is working properly.